Akhil Kumar

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Education

Indian Institute of Technology (IIT) Delhi — New Delhi, India
Bachelor of Technology in Chemical Engineering — *2016 — 2020*

Research Experience

Associate Computational Researcher — Apr 2023 — Present
Tsankov Lab, Icahn School of Medicine at Mount Sinai — *New York, NY*
Advisor: Alexander Tsankov (PI)

- Led a project stratifying microsatellite-stable colorectal cancer using bulk, single-cell, and spatial transcriptomics
- Co-first author on 2 publications and 1 manuscript in preparation; co-mentored 2 rotation PhD students
- Defined 9 molecular subtypes from 425 TCGA tumors, including a fetal-like regenerative subtype with worse survival
- Built a ~856k-cell scRNA-seq atlas of 294 tumors from 16 studies, benchmarking data integration with scib-metrics
- Built a Xenium spatial pipeline for ~2.6M cells (90 patients), benchmarking cell segmentation and admixture removal
- Compared subtypes by cell composition, differentiation potency, cell–cell signaling, gene regulation, and spatial niches
- Showed metaplasia converges on gastric goblet programs in 6 single-cell datasets, supporting a shared GI precursor
- Validated single-cell findings on TP53-associated tumor remodeling in lung adenocarcinoma across 2 bulk cohorts

Research Fellow — Feb 2021 — Nov 2022
Perumal Lab, IIT Delhi — *New Delhi, India*
Advisors: Perumal Vivekanandan (PI) and Manoj B. Menon (co-advisor)

- Led the lab's computational work; refactored its codebases and introduced reproducible, test-driven practices
- First or co-first author on 3 publications and 1 manuscript in preparation; guided a PhD student in data analysis
- Quantified temporal shifts in CpG depletion across 1.4M+ SARS-CoV-2 genomes, consistent with host adaptation
- Extended this analysis to 2,600+ HCV genomes (1977–2021), linking CpG loss to depletion of ZAP-binding motifs
- Mapped 11 stretches (≥100 nt, ≥99.9% conserved) across 6M+ SARS-CoV-2 genomes; 6 overlap known RNA structures
- Built a reproducible HPC pipeline to call mutations in 15.5M SARS-CoV-2 genomes and showed that early recurrence predicts predominance: 79–85% of early mutations reaching ≥1% frequency recurred within 28 days of first detection

Publications (First or Co-first Author)

1. 2025, Nature Cancer: Zhao, W.†, Nguyen, T. T.†, Bhagwat, A.†, **Kumar, A.**†, Giotti, B.†, …, Tsankov, A. M.* A cellular and spatial atlas of TP53-associated tissue remodeling defines a multicellular tumor ecosystem in lung adenocarcinoma.
2. 2025, Nature Communications: Bairakdar, M. D.†, Lee, W.†, Giotti, B.†, **Kumar, A.**†, Stancl, P.†, …, Tsankov, A. M.* Learning the cellular origins across cancers using single-cell chromatin landscapes.
3. 2024, Briefings in Functional Genomics: **Kumar, A.**, Kaushal, R., Sharma, H., Sharma, K., Menon, M. B.*, Vivekanandan, P.* Mapping of long stretches of highly conserved sequences in over 6 million SARS-CoV-2 genomes.
4. 2023, Pathogens: Mukherjee, S.†, **Kumar, A.**†, Samal, J., Gupta, E., Vivekanandan, P.*, Menon, M. B.* Selective depletion of ZAP-binding CpG motifs in HCV evolution.
5. 2022, Molecular Biology and Evolution: **Kumar, A.**†, Goyal, N.†, Saranathan, N., Dhamija, S., Saraswat, S., Menon, M. B.*, Vivekanandan, P.* The slowing rate of CpG depletion in SARS-CoV-2 genomes is consistent with adaptations to the human host.
6. In preparation: **Kumar, A.**†, Kaushal, R.†, Menon, M. B.*, Vivekanandan, P.* Early recurrence of mutations in SARS-CoV-2 predicts predominant mutations. Code

† Equal contribution * Recommendation letter writer

Peer Review

- Ad hoc reviewer, 1 manuscript each (2026): *Biology of Sex Differences* and *Discover Oncology*

Awards and Honors

2019 Award for outstanding contributions to the Student Affairs Council, the apex student body of IIT Delhi
2016 All-India Rank 1580 in JEE Advanced, the IIT entrance exam (~1.2 million candidates at the JEE Main stage)
2016 Rank 4 of 15,000 in the Jammu and Kashmir Engineering Entrance Examination; top scorer in Chemistry
2015 Kendriya Vidyalaya recognition for best all-round performance in a graduating class of 300
2014 State Rank 2 (Physics) and 3 (Biology), National Standard Examinations, IAPT
2012–13 Winner, regional and sub-regional quizzing and debating contests, KVS Social Science Exhibition

Undergraduate Research

Bachelor's Thesis — Fall 2019
Multiscale Modeling Group, IIT Delhi — *New Delhi, India*

- Contributed to the *in silico* design of small-molecule ligands to inhibit early-stage nucleation of insulin aggregation
- Identified EGCG and polyoxometalates as candidate ligands through literature review and computational modeling
- Docked EGCG into a partially folded insulin intermediate using ROSIE, the Rosetta modeling web server

Independent Study Fall 2019
Perumal Lab, IIT Delhi *New Delhi, India*
- Curated and filtered 889 human mtDNA sequences spanning ~50,000 years from the Ancient mtDNA Database
- Tracked shifts in mono- and dinucleotide composition and observed/expected ratios across time-binned samples
- Found significant shifts in TT, AG, and CT dinucleotide ratios over time and opposing purine trends on the two strands

Research Intern Summer 2019
Biomolecular Computation Group, Indian Institute of Science *Bangalore, India*
- Compared kinetic rate laws (mass-action, Michaelis–Menten, convenience kinetics) for dynamic metabolic models
- Built a convenience-kinetics model of glycolysis, the TCA cycle, and the respiratory chain in COPASI
- Simulated the model (92 reactions, 110 species, 4 compartments) and ran parameter sensitivity analysis

Research Intern Summer 2018
Srinivas Group, University of Oxford *Oxford, UK*
- Processed raw Smart-seq2 data from wild-type and *Aspp2* mutant early mouse embryos through Salmon quantification
- Applied 6-metric cell QC (498 high-quality cells retained), scran normalization, and selected 1,983 highly variable genes
- Resolved epiblast, mesoderm, visceral, and anterior visceral endoderm populations via clustering and marker genes

Teaching, Mentorship, and Leadership

Bioinformatics Tutor Summers 2023 — 2025
Center for Excellence in Youth Education, Icahn School of Medicine at Mount Sinai *New York, NY*
- Mentored 10 high-school students over 3 summers in an introductory bioinformatics program on cancer genomics
- Taught Python-based bioinformatics workflows on breast cancer transcriptomic datasets and their biological context
- Guided students in interpreting results, reading the literature, and presenting their findings

Student Mentor Jul 2019 — May 2020
Board for Student Welfare, IIT Delhi *New Delhi, India*
- Selected from 950 applicants for a 145+ member team supporting first-year students' transition to college life
- Mentored 5 first-year students through orientation and 4 academic guidance sessions per semester

Student Representative Apr 2018 — Apr 2019
Student Affairs Council, IIT Delhi *New Delhi, India*
- Served as 1 of 2 institute-wide representatives of 8,000+ students to the Senate of Deans and Associate Deans
- Conceived and launched a research mentorship program pairing undergraduates with postgraduate researchers

International Affairs Coordinator Apr 2018 — Apr 2019
Alumni Affairs & International Programs, IIT Delhi *New Delhi, India*
- Led a team of 6 in MoU outreach; helped revive 10 inactive MoUs by liaising across institute administrative divisions
- Revamped the buddy program for 30+ exchange students (3:1 peer mentorship) and organized 5 integration events
- Co-led a 16-member team hosting the Dalai Lama's campus visit, coordinating with Delhi Police and campus security
- Worked with the alumni association to organize silver and golden jubilee reunions for the 25- and 50-year classes

Tennis Captain (2018–19); Vice-Captain (2017–18) Apr 2017 — Apr 2019
Karakoram House, IIT Delhi *New Delhi, India*
- Led the house team to the semifinals of the institute tennis championship; organized 2 intra-house tournaments
- Ran 7 group and individual sessions to bring in new players from non-tennis backgrounds

Science Communication and Service

- Selected to tell a personal science story at a Story Collider show co-hosted with Mount Sinai (2025)
- Designed, 3D-printed, and donated a tactile Scrabble game for blind students; taught gameplay in a demo session
- Digitized an NGO's complete child-welfare database under committee supervision
- Delivered 8 weeks of educational and personal-development sessions for children at an orphanage

Skills and Interests

Programming	Python (scverse, pandas, NumPy, scikit-learn, matplotlib), R (Seurat, tidyverse, Bioconductor), Bash
Transcriptomics	Single-cell, bulk, and spatial (Xenium) transcriptomics analysis
Viral genomics	Sequence alignment (MAFFT), mutation calling (Nextclade), RNA structure prediction (MXfold2)
Reproducibility	Nextflow, Git/GitHub, Apptainer/Docker, conda/mamba/pixi, Jupyter, R Markdown
Computing	HPC (LSF), large-scale parallel processing
Interests	Ultramarathons (50-mile races; team Fastest Known Time, Staten Island perimeter, 2024), biking, chess, tennis, badminton, cricket (Mount Sinai team)